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AE-CFAR Multi-Target Detection for 24GHz Cascaded Automotive mmWave Radar

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Abstract

This paper addresses the need for multi-target detection using automotive mmWave radar in complex traffic scenarios and proposes a 24 GHz cascaded system and an AE-CFAR detection link in the RDM domain. The system employs SRK2401 + SRK2402 to construct a 3T8R cascaded transceiver front-end, with 8Rx parallel IF sampling on the receiving side to form a data cube. On the end side, STM32H753 is responsible for front-end configuration, timing scheduling, and data transfer, while TMS320C6748 performs range FFT, velocity FFT to construct the RDM, and executes AE-CFAR detection. The algorithm suppresses training window contamination through truncated mean robust noise estimation in the RDM domain, combines edge-aware small-side selection in the

range dimension for adaptive threshold correction, and utilizes 3T8R observation redundancy for cross-channel/cross-slot consistency screening to reduce false alarms caused by mutual interference pseudo-peaks and occasional noise. Simulation, anechoic chamber, and real-vehicle test results show that the system covers a detection range of 6–140 m and a speed measurement range of 14–225 km/h, with a range resolution of ±5 cm and a velocity resolution of ±0.7 km/h, supporting simultaneous tracking of up to 50 dynamic targets. In comprehensive scenarios, the false alarm rate is reduced by approximately 20%, the detection accuracy is improved by about 15%, the miss rate in multi-target working conditions at 150 m is reduced by more than 10%, and the end-side processing delay does not exceed 5 ms.

Keywords: 24GHz mmWave radar, cascaded MIMO, AE-CFAR, multi-target detection, edge-side deployment

1 Introduction

Automotive mmWave radar plays a crucial role in measurement within advanced driver assistance systems (ADAS) and autonomous driving perception links. It can stably provide distance, speed, and angle information even under visually degraded conditions such as rain, fog, low light, and backlighting. Therefore, it is widely used in core tasks such as multi-target detection, target list generation, and trajectory tracking [1–4]. However, real-world road scenes often feature dense traffic flow, a large dynamic range between strong and weak targets, significant occlusion and multipath effects, and frequent radar-to-radar interference. These factors can introduce structural false peaks or energy diffusion in the range-Doppler map, leading to biased detection threshold estimation, increased false alarms, and masking of weak targets. This reduces the stability of target lists and amplifies subsequent tracking errors [5–9].

At the system architecture level, enhancing the spatial observation dimension is an important approach to alleviate the insufficient resolvability of multiple targets. Time-division multiplexed (TDM) MIMO FMCW radar expands the equivalent aperture by constructing a virtual array, which can improve angular resolution and angle estimation accuracy with a limited number of physical antennas [10, 11]. Furthermore, cascading multiple transceiver chips can expand the number of TX/RX channels while keeping the complexity of a single chip manageable, forming a larger-scale virtual array. This provides higher degrees of freedom for multi-target separation and 3D positioning [12–15]. Meanwhile, to meet engineering requirements for higher-density point clouds and more stable target lists, cascaded arrays and signal processing links must be co-designed. In particular, a reproducible system implementation process must be established for reference clock sharing, deterministic triggering, and cross-chip amplitude-phase and time-delay calibration. This avoids elevated angular sidelobes and target splitting caused by insufficient coherence between channels [10, 12, 13].

At the detection algorithm level, constant false alarm rate (CFAR) threshold adaptation is a fundamental step for automotive radar to stably extract candidate point clouds from range-Doppler spectra. However, dense multi-targets and non-uniform

backgrounds can cause reference window contamination and edge effects, making it difficult for traditional statistical thresholds to balance detection rate and false alarm rate under different traffic conditions [16–19]. Research in the past five years has enhanced the robustness of CFAR from multiple directions: one category of work, targeting the engineering constraints of traffic-oriented mmWave radar, improves threshold stability in complex scenarios by modifying reference window sampling strategies or introducing environment discrimination mechanisms [16, 17, 19]; another category focuses on clutter non-uniformity and weak target detection, incorporating higher-order statistics, distribution adaptation, or multi-feature fusion into the CFAR framework to improve threshold estimation consistency in edge environments or under multi-target interference [20, 21]; additional studies integrate the point cloud generation process with temporal/spatial modeling of CFAR to enhance reflection point detection capability and downstream task usability in sparse point cloud scenarios [22]. Meanwhile, detection approaches based on representation learning methods such as autoencoders have also been used to replace or enhance traditional CFAR, improving structured target extraction under multi-target conditions [23].

It is worth emphasizing that automotive radar in road environments also faces issues of "false peaks" and "ghost targets" caused by mutual interference. Relying solely on CFAR with fixed thresholds or a single statistical hypothesis often misdetects residual interference as real targets, thereby increasing false alarms and disrupting tracking continuity [5–9]. Therefore, it is necessary to co-design "angular domain separation capability brought by cascaded arrays" and "adaptive CFAR threshold strategies for multi-targets and edge effects": the former improves the upper limit of multi-target resolvability, while the latter ensures stable target candidate sets in the presence of non-uniform backgrounds, reference window contamination, and interference, thereby providing high-quality inputs for clustering and tracking [24, 25]. Based on these ideas, this paper focuses on multi-target detection using 24 GHz cascaded automotive mmWave radar, conducting research on coherent acquisition under a cascaded architecture and RDM-domain threshold adaptation mechanisms. Incorporating recent advancements in CFAR improvement and interference suppression, it analyzes the synergistic effect of cascaded observation redundancy and the AE-CFAR threshold adaptation mechanism to suppress interference-induced false alarms and enhance detection stability of dense multi-targets in complex traffic environments. To achieve this goal, the work of this paper is summarized in three points: constructing a 3T8R cascaded transceiver front-end based on SRK2401 and SRK2402, with end-side pipelined processing implemented using MCU and DSP; proposing an RDM-domain AE-CFAR threshold mechanism for non-uniform backgrounds and multi-target contamination, incorporating robust noise estimation and edge-aware correction, and adopting tabulated thresholds to meet real-time requirements; utilizing 3T8R observation redundancy for cross-channel/cross-slot consistency screening, and verifying its effectiveness through simulation, anechoic chamber, and real-vehicle tests, presenting metrics such as false alarms, accuracy, missed detections, and time delay.

2 System Hardware Architecture and Cascaded MIMO Implementation

2.1 Overall Hardware Architecture and Data Link

The system adopts the 24 GHz FMCW system, with two radar transceiver chips SRK2401 and SRK2402 forming a cascaded transceiver front-end, supplemented by multi-channel synchronous sampling and cooperative baseband processing of MCU (STM32H753)/DSP (TMS320C6748), to achieve 3T8R channel expansion and RDM domain detection output. Hardware control and baseband processing are undertaken by MCU and DSP respectively: MCU is responsible for front-end configuration, timing scheduling and data transfer, while DSP is responsible for high-throughput FFT and detection calculations. Figure 1 shows the hardware architecture and data link: the reference clock and trigger link provide a unified time base for the two chips; the transceiver front-end completes FMCW ramp generation, transmission time slot switching and echo down-conversion; 8 channels of received IF enter multi-channel synchronous sampling after analog conditioning, forming a data cube organized by frame, time slot and channel; the baseband side completes the range FFT, velocity FFT and AE-CFAR detection modules, outputs candidate points and target lists, and releases them externally through the vehicle-mounted interface.

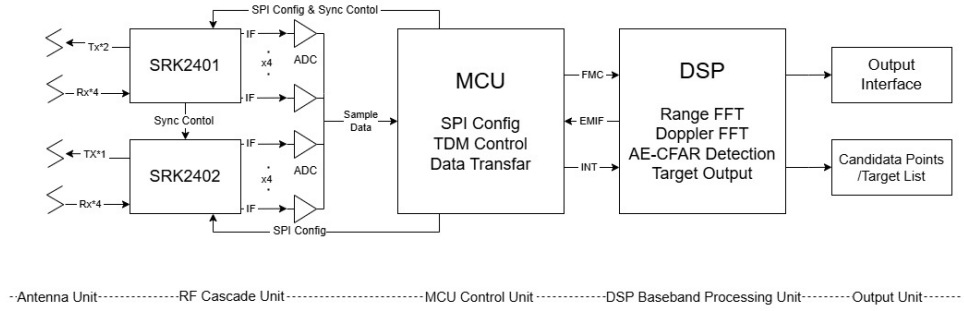


Fig. 1 Hardware Architecture and Data Link of the Cascaded System

The frequency sweep range, period and slope of the FMCW modulated signal determine the frequency mapping relationship of the range/velocity spectrum, which is the basis for data link parameter configuration. The system uses a 24 GHz band linear frequency-modulated chirp, with the modulated signal frequency sweep range of 24156.852–24907.891 MHz, a single period of approximately 250 μs, a frequency sweep bandwidth of approximately 58.90 MHz, corresponding to a frequency modulation slope of approximately 0.2358 MHz/μs. The above parameters are used to determine the calibration coefficient for range calculation, and provide constraints for the point configuration, window function and leakage control of range FFT and velocity FFT.

The echo at the receiving end is down-converted on the chip to form an IF beat signal. The system implements parallel sampling of 8 receiving channels in hardware, with a single-channel IF frequency range of 0–10 MHz. This IF bandwidth and amplitude range are used to determine the sampling rate, anti-aliasing filtering and range configuration of analog conditioning and multi-channel ADC; during the debugging phase, some channels can be observed to complete link tuning, while in the online processing phase, the RDM is constructed with full-channel sampling data and detection results are output.

2.2 Implementation of Cascade Transceiver Front-end with SRK2401 and SRK2402

Both SRK2401 and SRK2402 are 2T4R integrated FMCW radar transceivers, providing two transmit links and four receive links respectively. The system combines the receive links of the two chips in parallel to form 8 receive channels (8Rx). On the transmit side, time-division multiplexing (TDM) scheduling is adopted to form 3 effective TXs: TX1/TX2 of SRK2401 and TX1 of SRK2402 are selected, and they are transmitted in a three-slot polling manner within a frame, ensuring that only one TX is turned on at any time. This strategy reduces the impact of on-chip and inter-chip transmit crosstalk on the receive links on the hardware side, and forms a time-slot separable virtual array observation on the data side, providing stable input for subsequent angular domain processing and AE-CFAR detection[26].

To ensure the coherence of cross-chip channels, the two chips share the same reference clock to unify the frequency synthesis time base. The reference input of SRK2401 is 20 MHz[26]. The system distributes a single low-jitter 20 MHz reference source to the reference terminals of the two chips in a star topology, and performs equal-length and impedance control on the reference distribution traces to reduce phase drift and temperature drift differences introduced by the reference path. At the same time, monitoring signals such as DIVOUT are led out for lock status and online diagnosis of the clock link[26]. Frame timing alignment is achieved through deterministic triggering: under a unified time base, the two chips are synchronously enabled to enter the working state, and the frame controller switches the transmit channels according to the TDM table to ensure that the cross-chip sampling start point is consistent with the chirp timing.

In terms of interface and acquisition, each of the two chips provides I/Q differential IF outputs for four receive links, with an IF frequency range of 0–10 MHz[26]. The system sends 8 channels of IF in parallel to the analog conditioning and multi-channel synchronous ADC, forming a data cube organized by "fast time (range dimension) + slow time (velocity dimension) + channel dimension (8Rx) + slot dimension (3TX polling)", providing a unified data structure for the baseband-side FFT and detection links.

2.3 MCU and DSP Collaboration and Baseband Processing Interface

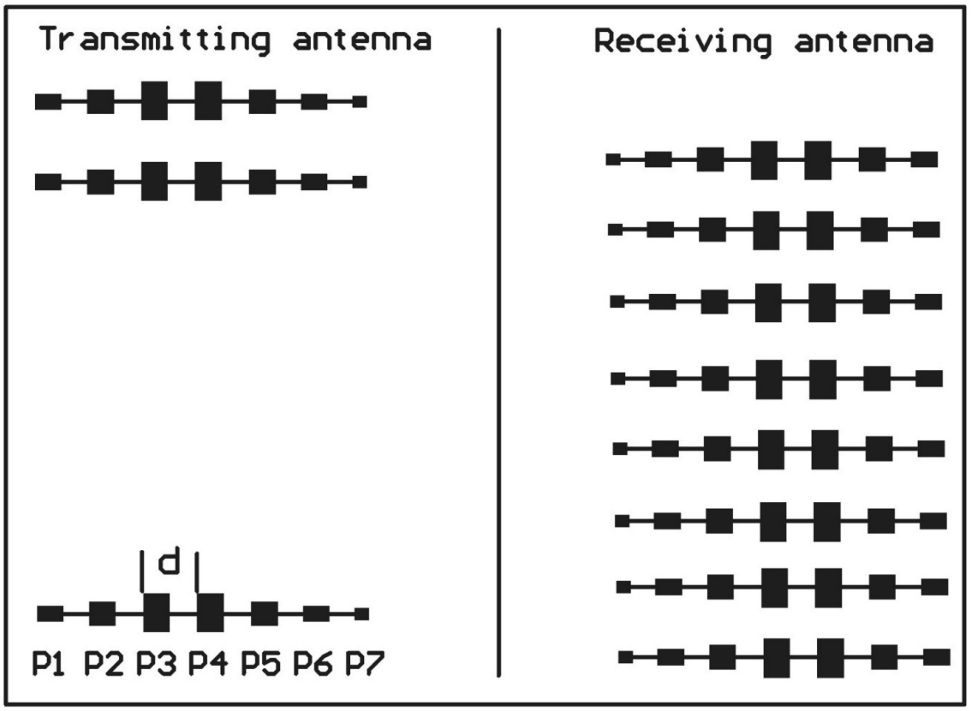
The system adopts a collaborative division-of-labor architecture between MCU and DSP to balance timing determinism and computational throughput. The MCU is responsible for the SPI configuration and state management of SRK2401 and SRK2402 (such as ramp parameters, channel enabling, gain levels, etc.), frame triggering and TDM time slot scheduling, sampling link control and data cache management, as well as data packaging and release for external interfaces. The DSP is in charge of high-throughput baseband processing: performing range FFT and velocity FFT on the data of each channel to construct a range-velocity two-dimensional spectrum (RDM), and executing AE-CFAR detection in the RDM domain to output candidate points and target candidate lists.

To ensure real-time performance on the edge side, the data link is implemented in a pipelined manner. On the MCU side, multi-channel sampling data is written into a ring buffer via DMA, with index management carried out by frame and time slot; subsequently, the data is sent to the DSP side in block transmission mode for FFT and detection calculations. The detection output adopts a compact structure (range bin, velocity bin, amplitude, channel/time slot index) for use by subsequent target management and tracking modules. This division of labor and pipelining mechanism can maintain stable processing latency under the condition of increased data scale caused by cascaded channel expansion, and provide a continuous and stable set of target candidates for multi-target detection in complex traffic scenarios.

2.4 Implementation and Acceptance of Cascaded MIMO Antenna

The system adopts a 3T8R cascaded array antenna to match the channel organization of two-chip cascading. The array uses a microstrip linear array series-fed structure, and reduces mutual interference between arrays through edge feeding excitation; the physical layout integrates 3 groups of transmitting units and 8 groups of receiving units on a small-sized substrate to meet vehicle-mounted integration constraints. Figure 2 shows the array model and channel organization, which are used to clarify the layout relationship of transmitting/receiving arrays, the organization of feeding traces, and the setting of array spacing [27].

The center frequency of the array is aligned to 24.1 GHz and covers the operating bandwidth of 23.975–24.225 GHz. The in-band reflection coefficient meets the engineering constraint of $S_{11} \leq -10\,\mathrm{dB}$; the radiation pattern indicators focus on main lobe gain and sidelobe suppression, which are used to support the acquisition of medium and long-distance echo energy and reduce the interference of spurious echoes from non-main lobe directions on threshold estimation. Figure 3 shows the array gain and radiation angle, which is used to verify the relationship between main lobe pointing and forward coverage; Figure 4 shows the array operating frequency and bandwidth, which is used to verify the alignment of center frequency and operating bandwidth [27].



	P1	P2	P3	P4	P5	P6	P7
Width	2.1mm	3.85mm	4.94mm	4.94mm	3.85mm	2.1mm	1.7mm
long	3.85mm	3.85mm	3.85mm	3.85mm	3.85mm	3.85mm	2.1mm

Fig. 2 Cascaded MIMO Array Model and Channel Organization

After the completion of the array prototype manufacturing, the tests include frequency band scanning and far-field pattern measurement: the former covers the operating bandwidth and measures the response point by point to verify the in-band matching and bandwidth coverage; the latter records the gain at different angles through a rotating platform to evaluate the main lobe width and sidelobe level. Figure 5 shows the physical array and test configuration, which is used to illustrate the engineering implementation of instrument connection, test environment, and measurement process [27].

2.5 Interface Definition from Acquisition to Detection

Multi-channel sampling data constructs the Range-Doppler Map (RDM) on the baseband side in the order of "range FFT + velocity FFT", and the RDM serves as the unified input domain for the AE-CFAR detection module. Figure 6 shows a schematic diagram of the processing chain, which is used to clarify the data flow from ADC

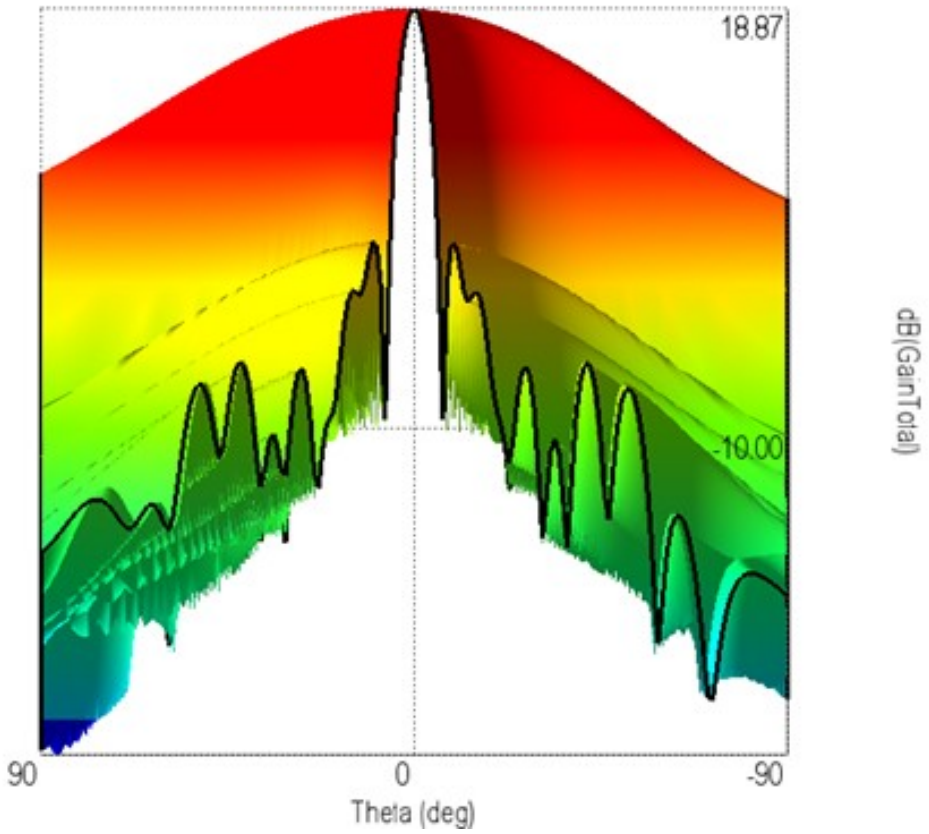


Fig. 3 Array Gain and Radiation Angle

sampling, range/velocity FFT to detection and range-velocity calculation. This interface definition ensures that the observation gain brought by the expansion of cascaded array channels can be effectively utilized in the unified RDM representation, and provides structured input for the subsequent threshold estimation of AE-CFAR under conditions of dense multi-targets and non-uniform backgrounds.

2.6 Implementation Constraints: Linearity Budget and Calibration Maintenance

To ensure the stable operation of the cascaded chain in complex traffic scenarios, the system constrains both linearity and channel consistency in implementation. The SRK2401 receiver chain has a typical conversion gain of approximately 29 dB, a noise figure of approximately 9 dB, and an input compression point of approximately -8 dBm [26]. Therefore, the system performs input power budgeting and operating point setting for transmit power, antenna isolation, and short-range strong reflections to prevent the receiver chain from entering the compression region, which would cause

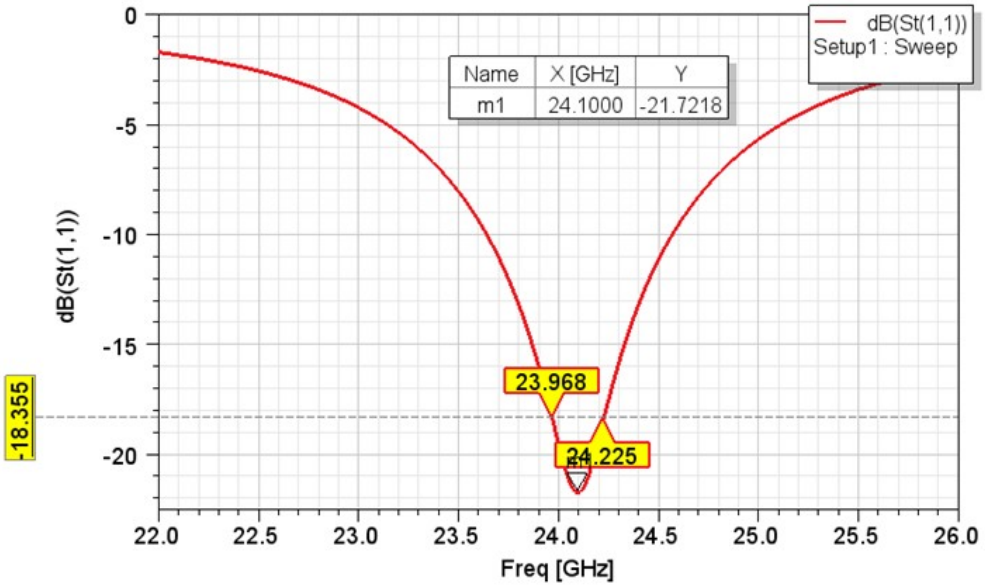


Fig. 4 Array Operating Frequency and Bandwidth

RDM energy diffusion and false alarm elevation. On the other hand, the system performs amplitude-phase and group delay calibration on the 8 receiving channels, and establishes a grouped calibration table according to TDM time slots to compensate for inconsistencies introduced by wiring differences, channel differences, and temperature drift, ensuring the stability of virtual array reconstruction and the consistency of angular domain output.

3 Design of AE-CFAR Detection Algorithm for 3T8R Cascaded System

3.1 Overall Idea and Division of Labor Principles

The system adopts a 3T8R cascaded system: 3 TDM transmission time slots and 8 receiving channels for parallel sampling. While this structure improves the spatial observation dimension, it also significantly increases the challenges of end-side data throughput and detection stability[5–9]. To obtain a continuous and stable set of target candidates in complex traffic scenarios, this paper designs the detection link as a series structure of "RDM construction – AE-CFAR threshold detection – 3T8R consistency fusion" and embeds it in the MCU and DSP collaborative framework: the MCU is responsible for timing determinism, continuous data flow, and task scheduling, while the DSP is responsible for high-throughput calculations such as two-level FFT and threshold detection, thereby maintaining stable processing delay and suppressing false alarms caused by residual mutual interference under limited computing power and bandwidth conditions.

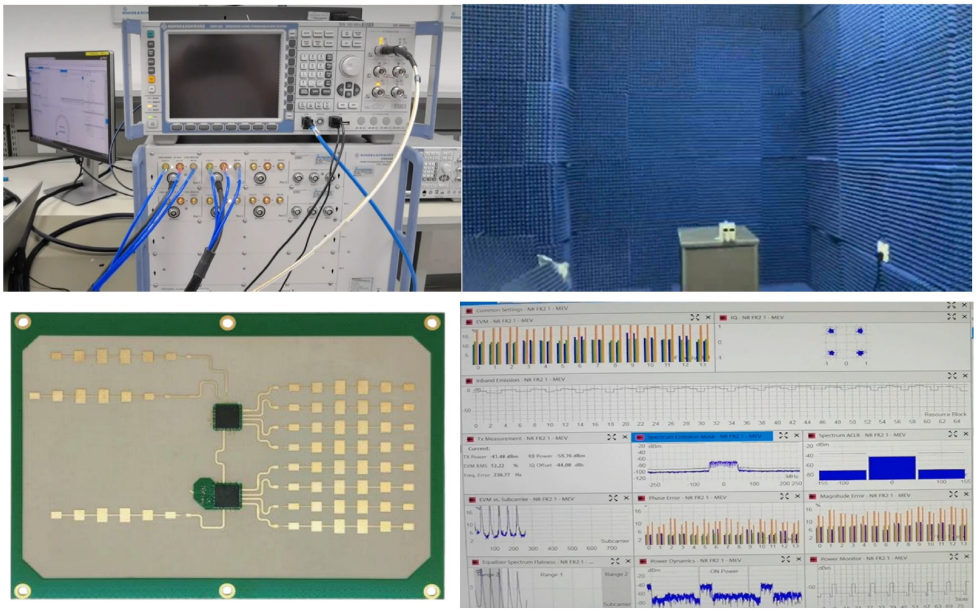


Fig. 5 Array Physical Object and Test Configuration

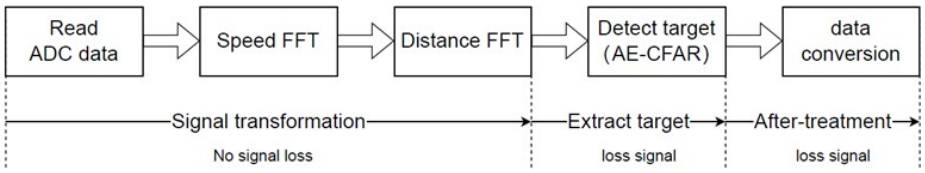


Fig. 6 Processing Chain and Detection Interface

3.2 Data Structure and End-Side Processing Pipeline

3.2.1 3T8R Data Organization

Define the fast-time sampling of the s-th TDM transmission time slot ($s \in \{1, 2, 3\}$) and the r-th receiving channel ($r \in \{1, \ldots, 8\}$) as $x_{r,s}[n, m]$, where $n = 0, \ldots, N-1$ is the sampling point index of each chirp, and $m = 0, \ldots, M-1$ is the chirp sequence index. The end-side cache is organized in units of frames, using the index structure of `frame→slot→rx→chirp→samples`, so that data from different transmission time slots can be separated at the storage level, and it is convenient for the DSP to complete RDM construction and detection according to time slots.

3.2.2 MCU-Side Tasks: Timing Scheduling and Data Flow Management

The MCU side mainly undertakes tasks strongly coupled with hardware timing: completing SRK2401/SRK2402 register configuration and state management through SPI; switching transmission channels according to the three-time-slot TDM table within the frame and maintaining trigger timing; driving the multi-channel synchronous sampling link and maintaining the frame/slot/channel index of the ring buffer; using DMA to complete sampling block writing and block-level data transfer; receiving the candidate point list output by the DSP and completing timestamp alignment and interface release. The above tasks ensure the continuous output of 3T8R data flow within the frame and provide stable input for subsequent FFT and detection on the DSP side.

3.2.3 DSP-side Task: RDM Construction and Detection Calculation

The DSP side performs a two-level FFT on each (r, s) to construct a range-velocity two-dimensional spectrum (RDM), and executes AE-CFAR threshold detection in the RDM domain to obtain the candidate point set $\mathcal{D}_{r,s}$. Due to mutual interference residues and non-uniform backgrounds, stripes/false peaks and local clutter mutations are introduced in the RDM. If the threshold estimation is unstable, it will lead to a surge in false alarms and disrupt downstream processing [5, 6, 8, 9]. Therefore, the detection strategy needs to simultaneously possess the capabilities of reference window contamination suppression and clutter edge perception.

3.3 Construction of RDM: Range FFT and Velocity FFT

For the sampled data $x_{r,s}[n, m]$ of the s-th transmission slot and the r-th receiving channel, the range FFT is defined as

$$X_{r,s}[k, m] = \sum_{n=0}^{N-1} x_{r,s}[n, m] \exp\left(-j2\pi \frac{kn}{N}\right) \tag{1}$$

The velocity FFT is defined as

$$Z_{r,s}[k, \ell] = \sum_{m=0}^{M-1} X_{r,s}[k, m] \exp\left(-j2\pi \frac{\ell m}{M}\right) \tag{2}$$

where k is the range bin and ℓ is the Doppler bin. The detection input is the power spectrum $P_{r,s}[k, \ell] = |Z_{r,s}[k, \ell]|^2$, and threshold estimation and candidate point output are performed in this domain.

3.4 AE-CFAR Detection: Contamination Suppression and Edge-Aware Threshold Estimation

3.4.1 Definition of 2D Training Window

For the CUT position (k, ℓ), guard cells G_r, G_d and training cells T_r, T_d are set in the range dimension and velocity dimension, respectively. The training set is

$$\mathcal{W}(k,\ell) = \{(i,j) : |i-k| \le T_r + G_r,\ |j-\ell| \le T_d + G_d\} \setminus \mathcal{G}(k,\ell) \tag{3}$$

The guard region is

$$\mathcal{G}(k,\ell) = \{(i,j) : |i-k| \le G_r,\ |j-\ell| \le G_d\} \tag{4}$$

Considering that clutter edges in the vehicle forward scene more often appear along the range dimension, $\mathcal{W}$ is divided into left and right sub-windows along the range dimension:

$$\mathcal{W}_L = \{(i,j) \in \mathcal{W} : i < k\} \qquad \mathcal{W}_R = \{(i,j) \in \mathcal{W} : i > k\} \tag{5}$$

3.4.2 Contamination Suppression: Truncated Mean Noise Estimation

To reduce the threshold elevation caused by the contamination of the training window by strong targets, a sorted and culled truncated mean estimation is adopted. Let the set of training window power samples $\{P[i,j] : (i,j) \in \mathcal{W}\}$ be sorted in ascending order as $u_{(1)} \le \cdots \le u_{(Q)}$, where $Q = |\mathcal{W}|$. After culling the top K samples, the noise estimation is

$$\widehat{\sigma}^2_{\mathcal{W}} = \frac{1}{Q-K} \sum_{q=1}^{Q-K} u_{(q)} \tag{6}$$

This estimation can suppress the contamination of the reference window when multiple targets are dense and outliers exist, and maintains stable edge-side computational load through fixed windows and fixed culling ratios [18, 19].

3.4.3 Edge Discrimination: Small-Side Selection and Effective Unit Update

In edge scenarios, the inconsistent background statistics on both sides of the training window can lead to systematic bias in the threshold. The robust means μ_L, μ_R are calculated for the left and right sub-windows, respectively:

$$\mu_L = \mathrm{TrimMean}(\mathcal{W}_L) \qquad \mu_R = \mathrm{TrimMean}(\mathcal{W}_R) \tag{7}$$

And the edge index is defined as

$$E = \frac{|\mu_L - \mu_R|}{\min(\mu_L, \mu_R) + \epsilon} \tag{8}$$

When $E > \tau_E$, the low-background side is used as the source of noise estimation:

$$\widehat{\sigma}^2 = \begin{cases} \mu_L, & \mu_L < \mu_R \\ \mu_R, & \mu_R \le \mu_L \end{cases} \tag{9}$$

And the number of effective training units N_{eff} is updated to the number of samples after culling in the corresponding sub-window; when $E \le \tau_E$, the full-window estimation $\widehat{\sigma}^2 = \widehat{\sigma}^2_{\mathcal{W}}$ is adopted. This "edge discrimination + small-side selection" mechanism is used to alleviate the elevation effect of the high-energy side of the edge on the threshold, thereby reducing the risk of missed detection of weak targets [16, 17].

3.4.4 Threshold Generation and Mutual Interference Robustness

The threshold is generated as

$$T = \alpha(P_{\text{fa}}, N_{\text{eff}})\widehat{\sigma}^2 \tag{10}$$

where P_{fa} is the target false alarm rate. To adapt to real-time computing on the end-side, $\alpha(\cdot)$ is implemented using an offline Look-Up Table (LUT); when mutual interference residues are significant, a small margin can be introduced to α to suppress false alarms induced by streaks or false peaks [6, 8]. For each CUT, if $P_{r,s}[k,\ell] > T$, a candidate point is output, and its (k,ℓ), amplitude, and channel/slot indices are recorded to form the set $\mathcal{D}_{r,s}$ for subsequent fusion.

3.5 3T8R Consistency Fusion: False Alarm Suppression for Cascaded Structures

The cascaded system has redundant observations of 8Rx and 3 transmission slots within the same frame. Mutual interference false peaks and occasional noise peaks usually exhibit inconsistency across channels/slots, while real targets have stronger consistency in the range-velocity neighborhood. Based on this difference, this paper introduces consistency filtering at the candidate point level as a lightweight post-processing step after AE-CFAR.

Let $\mathcal{N}_{\Delta k,\Delta \ell}(k,\ell)$ be the neighborhood matching operator for (k,ℓ) (allowing offsets of Δk and $\Delta \ell$ in distance and Doppler, respectively). For a candidate coordinate (k,ℓ), the consistency count is defined as

$$C(k,\ell) = \sum_{s=1}^{3}\sum_{r=1}^{8} \mathbf{1}\Big(\exists (k',\ell') \in \mathcal{D}_{r,s},\ (k',\ell') \in \mathcal{N}_{\Delta k,\Delta \ell}(k,\ell)\Big) \tag{11}$$

where $\mathbf{1}(\cdot)$ is the indicator function. The candidate point is retained if $C(k,\ell) \ge \eta$; otherwise, it is judged as an inconsistent false alarm and rejected. The threshold η and neighborhood scales Δk, $\Delta \ell$ are set according to system resolution and scene intensity: a higher η is adopted to enhance suppression when mutual interference is significant or traffic is dense; η is reduced to improve detection rate when weak targets

are prioritized. The filtered candidate set is denoted as $\mathcal{D}$, which serves as the input for subsequent angular domain processing and target list generation.

3.6 Edge Implementation: Pipelined Organization and Complexity Control

The edge side adopts dual-buffer pipelined execution to ensure real-time performance. On the MCU side, DMA is used to write sampling blocks into a circular buffer, and directly indexable data blocks are established according to "time slot–channel–chirp group"; when the data block of a time slot is ready, block transmission is triggered to enter the calculation queue on the DSP side. The DSP side performs two-level FFT on the input block to generate RDM, and then scans the RDM with fixed window parameters to execute AE-CFAR; the sorting and elimination stage adopts the implementation of fixed Q and fixed K, and can use partial selection or approximation strategies to reduce the sorting cost to stabilize the calculation time. Candidate point consistency fusion is executed at the candidate point level, and its computational complexity is approximately linearly related to the number of candidate points, so stable processing load and output delay can be maintained during long-term operation on the edge side.

4 Simulation Experiments, Whole Machine Testing and Result Analysis

4.1 Experimental Design and Evaluation Indicators

This experiment evaluates the 3T8R cascaded system and the AE-CFAR detection link through two paths: simulation verification and whole machine actual measurement. The simulation part is used to quantify the stability of the threshold strategy under the conditions of dense multi-targets and enhanced noise; the whole machine test part is used to verify the closed-loop effectiveness from the radio frequency link, modulation and sampling to the edge real-time detection output; the system-level test further covers two typical application scenarios: darkroom-controlled multi-targets and open road complex environments.

The evaluation indicators include: (1) False alarm rate and miss rate, used to measure the stability of threshold estimation; (2) Detection accuracy, used to measure the consistency between the detection output and the real target; (3) Distance/speed measurement capability and resolution, used to measure measurement accuracy; (4) Multi-target concurrency capability, used to measure the availability in congested or dense traffic scenarios; (5) Edge processing delay, used to measure the feasibility of real-time deployment. The overall system performance goals and actual measurement verification indicators are: detection range 6–140 m, speed measurement range 14–225 km/h, distance resolution ± 5 cm, speed resolution ± 0.7 km/h, and support simultaneous tracking of up to 50 dynamic targets; at the same time, compared with other CFAR variants, AE-CFAR can reduce the false alarm rate by about 20% and improve the detection accuracy by about 15% in complex scenarios.

4.2 Simulation Verification: Performance of AE-CFAR Multi-Target Detection

4.2.1 Simulation Method and Parameter Configuration

The simulation was completed in the MATLAB environment. The ADC output was generated according to the actual radar signal processing chain, and a range-velocity two-dimensional spectrum (RDM) was constructed. The processing flow is as follows: perform range FFT and velocity FFT on the beat frequency signal to obtain range and velocity information, then input it into AE-CFAR for threshold detection and false alarm control. After detection, the range is calculated from the beat frequency and frequency modulation slope, and the velocity is calculated from the Doppler frequency shift. The simulation parameters consist of fixed number of sampling points N, number of chirps M, false alarm probability configuration, training/guard cell configuration, and combinations of target quantity and noise figure. The specific radar signal parameters are listed in Table 1. To intuitively show the impact of threshold detection on the quality of candidate points in the RDM domain, Figure 7 presents the comparison results before and after AE-CFAR processing.

Table 1 Simulated radar signal parameters (corresponding to Table 6)

Parameter Item	Description
N	Number of sampling points per chirp
M	Number of chirps (slow time length)
P_{fa}	Target false alarm probability
T_r, T_d	Training cells (range/velocity dimension)
G_r, G_d	Guard cells (range/velocity dimension)

4.2.2 Multi-Target Detection Results and Phenomenon Analysis

In the multi-target simulation, the program randomly generates different numbers of targets, each with random range, velocity, and signal strength, and noise is added to simulate the false alarm background. Figure 8 shows the range and velocity detection results under different target quantities, where the point clusters correspond to the distribution of detected targets; Figure 9 shows the corresponding signal strength distribution, with colors from dark to light representing echo amplitudes from weak to strong. The simulation results show that when the number of targets increases to 60, the number of detected targets remains consistent with the set number of targets, indicating that the detection chain still has good stability in outputting candidate points under dense target conditions.

From the perspective of the RDM domain, the most common problem when multiple targets are dense is that the training window is contaminated by multiple strong peaks simultaneously, leading to an overall increase in the threshold and causing weak target masking; secondly, the uneven local background causes sudden changes in the threshold at the edges, inducing false alarms. In Figure 9, both strong and weak echoes

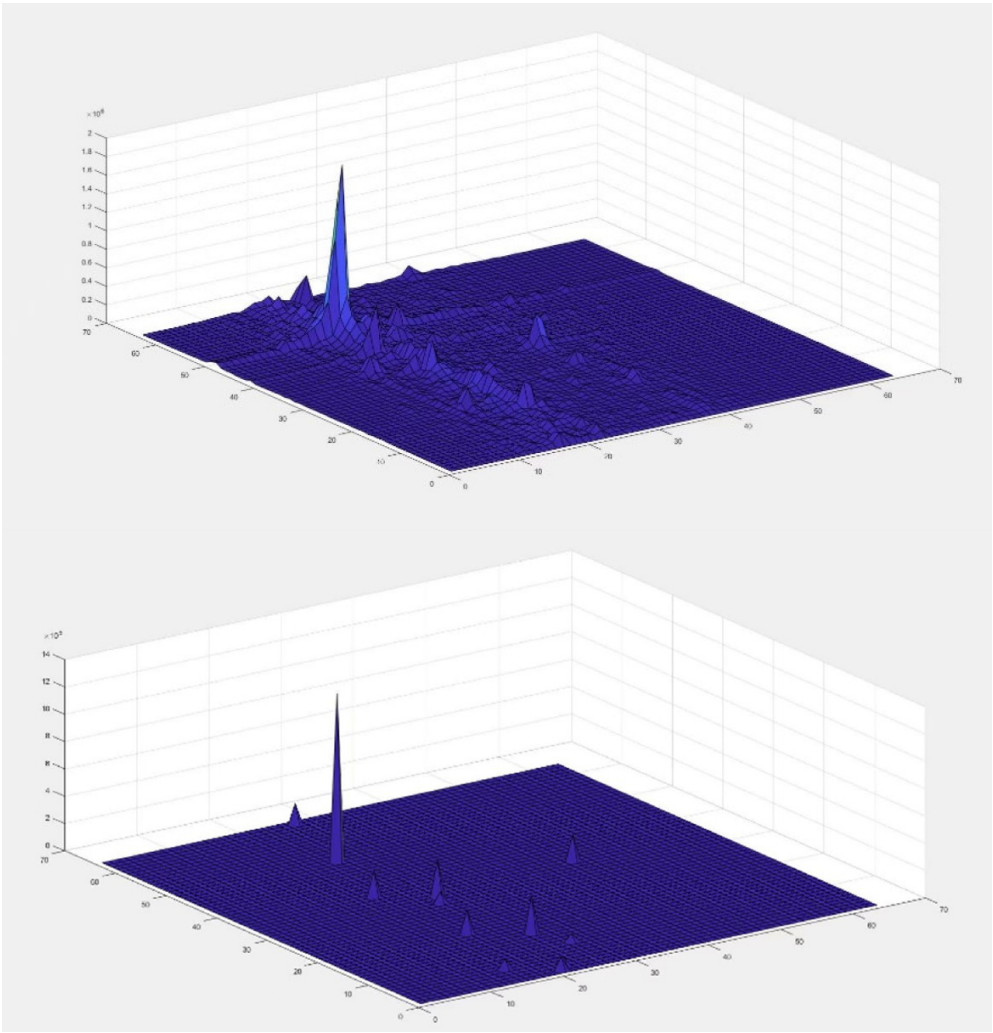


Fig. 7 Schematic diagram of RDM comparison before and after AE-CFAR processing

exist, providing a typical stress condition of "strong contamination + weak targets" for threshold estimation. AE-CFAR maintains a clear target cluster structure under this condition, indicating that its contamination suppression and edge perception mechanisms can suppress threshold drift at a statistical level, thereby avoiding systematic suppression of weak targets in the neighborhood of strong peaks.

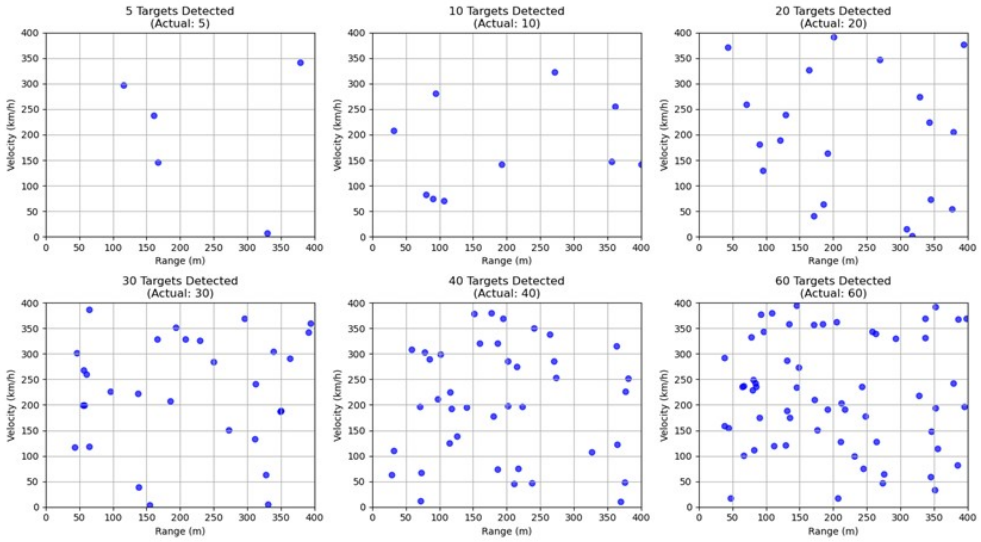


Fig. 8 Schematic diagram of range-velocity detection results under multi-target conditions

4.2.3 Comparison of Different CFAR Algorithms and Analysis of Threshold Stability

To compare the differences in threshold strategies, three scenarios were set: single target (noise figure 1.0), 5 targets (noise figure 1.5), and 10 targets (noise figure 2.0), and AE-CFAR, CA-CFAR, GO-CFAR, and SO-CFAR were implemented respectively. Figure 10 shows the comparison results, where the blue curve is the input signal, the threshold curves are the estimated thresholds of each method, and the detection markers are used to compare the actual target positions. Comparative analysis shows that under conditions of increased noise and multi-target interference, whether the threshold can avoid being raised by abnormal samples is the key factor determining missed detections and false alarms; AE-CFAR maintains more stable detection output by dynamically adjusting the threshold and noise estimation method, which can reduce the false alarm rate by about 20% and improve the detection accuracy by about 15% compared with other methods.

Furthermore, the behavioral differences of different methods can be explained from the shape of the threshold curve: CA-CFAR relies on average statistics and tends to shift upward as a whole with strong peak contamination; GO-CFAR tends to select a higher side background to suppress false alarms when there is local strong interference, but at the expense of weak target detection; SO-CFAR may have segments with excessively low thresholds in non-uniform backgrounds, leading to false alarms. AE-CFAR combines "contamination suppression (eliminating strong outliers)" and "edge perception (small side selection)", making the threshold closer to the effective background level in scenarios of strong contamination and edge mutations, thereby improving the trade-off between false alarms and missed detections.

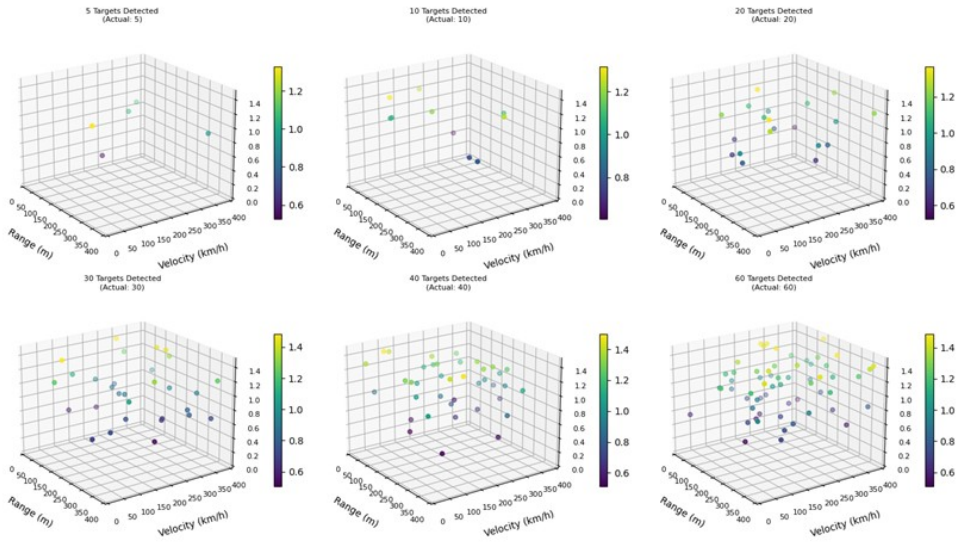


Fig. 9 Schematic diagram of signal strength distribution under multi-target conditions

4.3 System-Level Testing: Verification of RF Link, Modulation, and Sampling Consistency

4.3.1 Array Antenna Testing and Radiation Characteristics Acceptance

After the optimization of the array antenna, frequency band scanning and radiation pattern measurement are carried out in an anechoic chamber (electromagnetically shielded environment) to avoid the impact of external electromagnetic interference on the measurement results. The test uses a signal source, a spectrum analyzer, and a far-field test system, covering the 23.975–24.225 GHz frequency band to record antenna responses point by point, and measures gains in different directions under far-field conditions, focusing on indicators such as main lobe width and side lobe amplitude. Figure 11 shows a schematic of the array antenna measurement data. The significance of this step is as follows: whether the in-band matching and radiation pattern of the antenna meet the standards directly affects the echo energy and side lobe spurs, which in turn affects the RDM background statistics and the stability of threshold estimation; therefore, antenna acceptance is an indispensable pre-link in the "hardware-algorithm" closed loop.

4.3.2 Measurement and Linear Calibration of FMCW Modulated Signals

The transmitting end uses an MCU to generate a 24.1 GHz FMCW modulated signal, which is radiated by the array antenna after voltage following and RF power

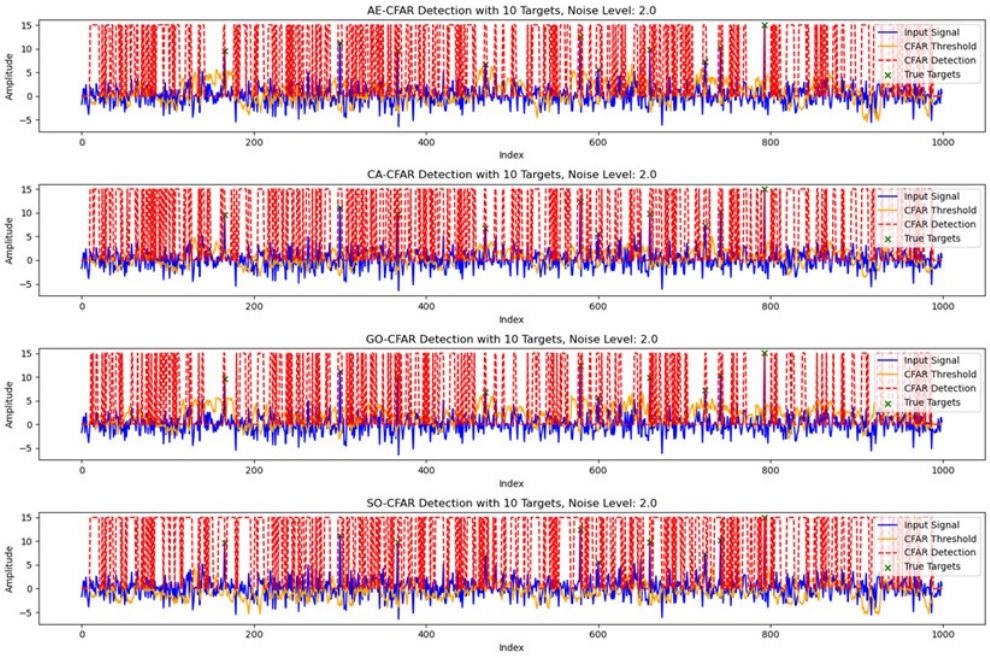


Fig. 10 Comparison of different CFAR methods: 10 targets, noise figure 2.0

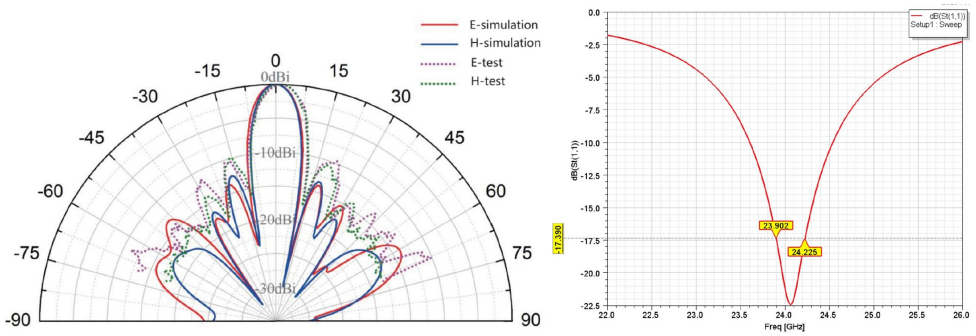


Fig. 11 Schematic of array antenna measurement data

amplification. A spectrum analyzer is used to measure the modulated signal to verify the frequency modulation linearity and timing parameters. Figure 12 shows the measurement results: the modulation frequency range is 24907.891–24156.852 MHz, the period is approximately 250 μs, the frequency span is 58.90 MHz, and the slope is approximately 0.2358 MHz/μs. The above parameters are used to align the distance calculation mapping relationship: if there is a drift in the slope or period, it will lead to the broadening of the range profile, peak offset, or velocity spectrum leakage, which in

turn will increase the statistical fluctuation of the training window and raise the risk of false alarms; therefore, this measurement is used to confirm the basic conditions for the subsequent RDM construction.

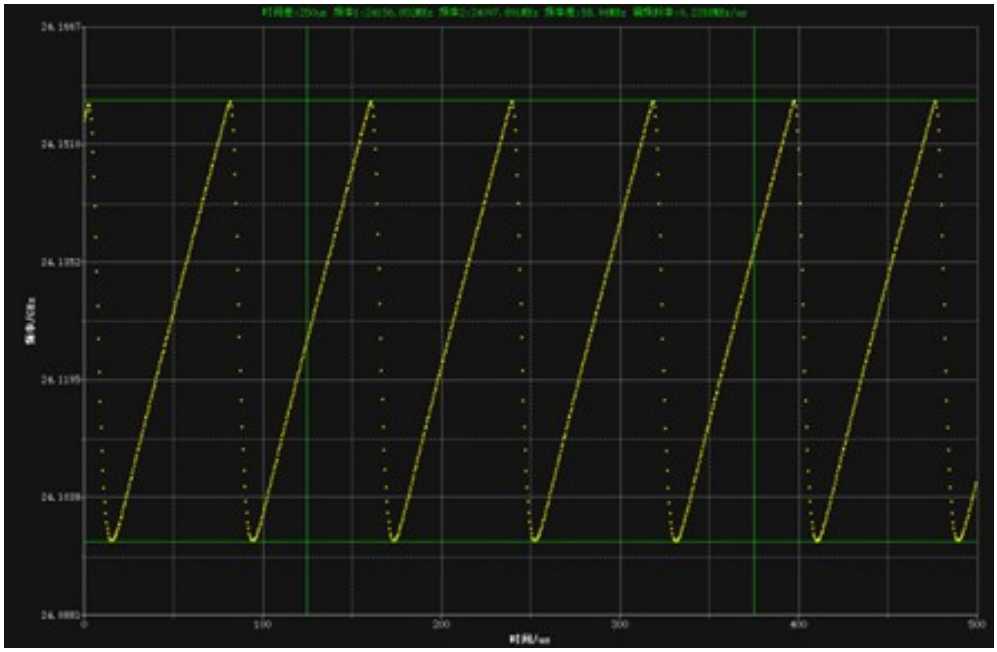


Fig. 12 Measurement results of FMCW modulated signals

4.3.3 IF Link Observation and Sampling Configuration Analysis

The receiving end downconverts to obtain the IF beat signal, and the quality of the IF signal determines the availability of the range and velocity spectrum. Figure 13 shows the measured waveform of the IF signal, which is used to check the amplitude range of the receiving link and the stability of the beat frequency. Since the system has 8 receiving channels but the test spectrum analyzer only supports 2-channel input, 2 channels are selected for observation during the debugging phase to complete the link health check and parameter setting. This limitation does not affect the full-channel input for end-side detection, but requires that in engineering implementation, the "channel sampling + statistical consistency" method is used to ensure that the 8 channels do not have significant mismatches or abnormal noise elevation; otherwise, a systematic deviation of "continuous inconsistency of local channels" will occur during the 3T8R consistency fusion phase, affecting the selection of the fusion threshold.

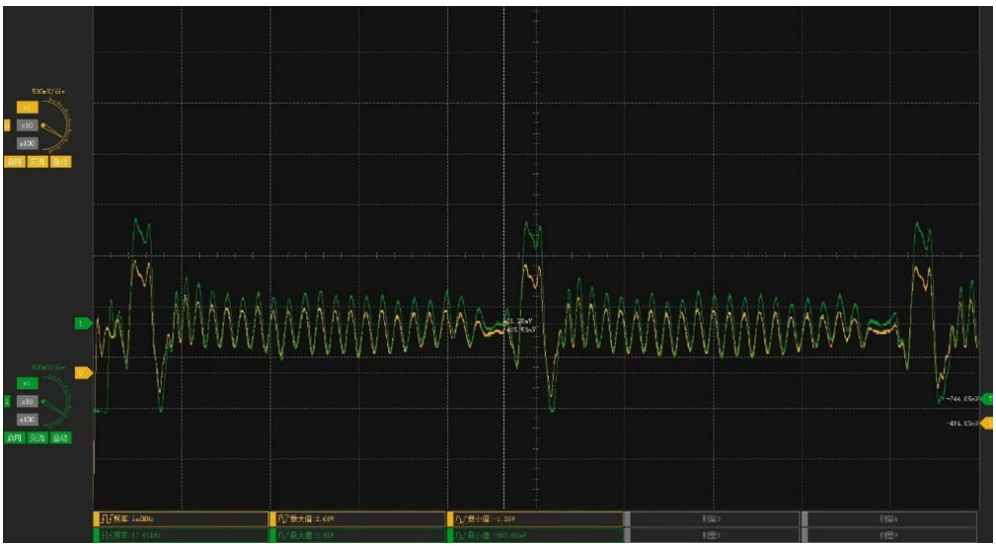


Fig. 13 Schematic of the measured waveform of the IF beat signal

4.4 Anechoic Chamber Test: Verification of Multi-Target Capability in a Controlled Electromagnetic Environment

4.4.1 Test Environment and Test Procedure

The anechoic chamber test is conducted in an electromagnetically shielded environment to reduce the impact of external electromagnetic interference on measurement results and evaluate the radar system performance under controllable conditions. The test environment and data are schematically shown in Figure 14. The test is divided into two parts: the first part uses PC software to control the test equipment to transmit simulated target signals, while collecting the target data output by the radar for analysis; the second part involves the radar actively transmitting signals, which are then received by the test software for data analysis. This design allows the test to evaluate both "the response consistency of the receiving link and detection algorithm under controlled inputs" and "the end-to-end output stability of the radar under active transmission conditions", thereby distinguishing algorithm performance from hardware link factors.

4.4.2 Test Configuration, Concurrent Scale, and Recording Method

In the receiving link verification phase, the test software displays echo peaks with distance as the horizontal axis and signal strength as the vertical axis, and provides a speed channel setting interface to list the measured relative speeds of the targets. Test equipment is used to transmit simulated target signals with speeds ranging from 0 to 250 km/h and distances from 0 to 200 m, covering various concurrent configurations such as 1, 5, 10, 20, 40, and 60 targets; the radar module processes the received data and outputs it as TTL data, which is parsed by PC software to present the target's

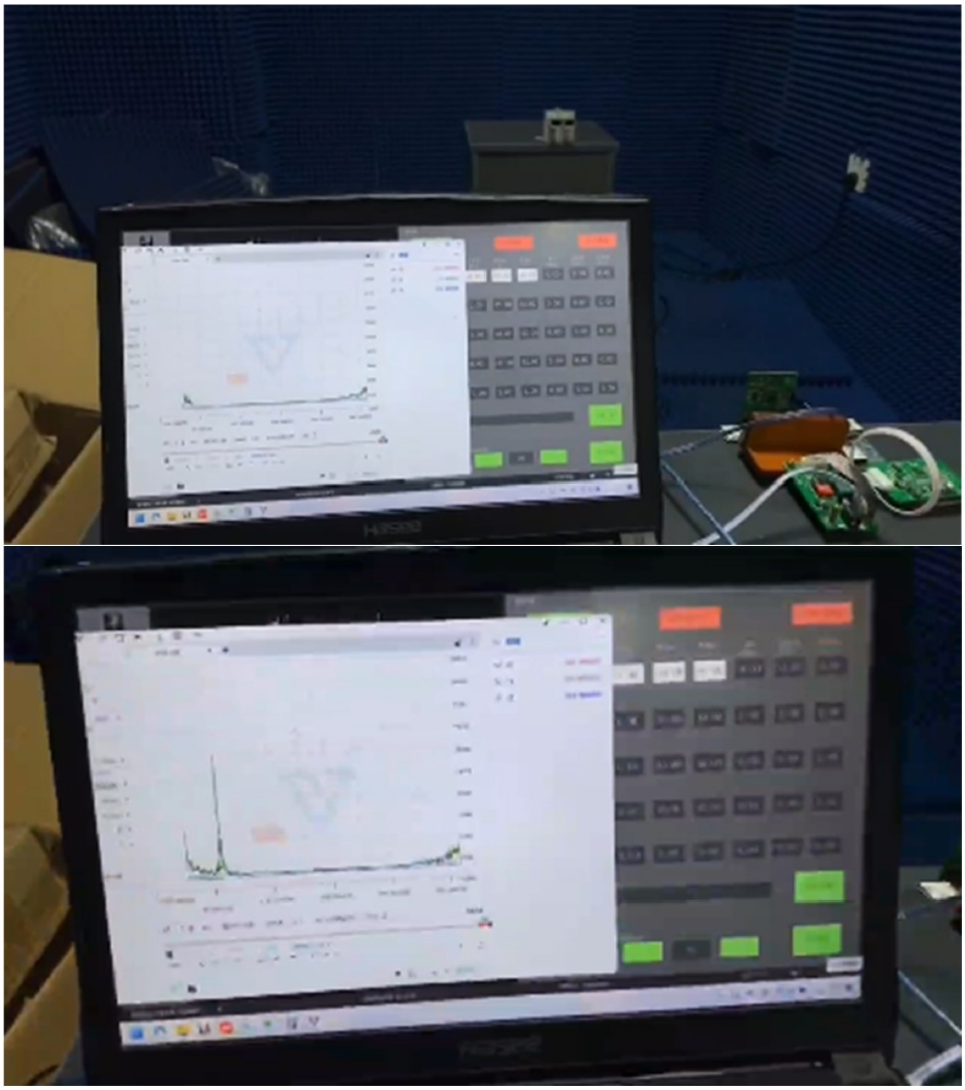


Fig. 14 Schematic diagram of performance testing in an anechoic chamber environment

speed and distance information. This configuration covers the complete pressure range "from sparse targets to dense targets", and can be used to observe the changing trends of training window contamination and threshold drift as the number of concurrent targets increases.

4.4.3 Anechoic Chamber Results and Analysis: Multi-Target Density, Threshold Stability, and Concurrent Upper Limit

Figure 15 shows an example of measured speed and distance under the condition of 60 targets. From the perspective of result characteristics, under dense target conditions, a large number of peaks appear simultaneously in the distance and speed channels, which is a working condition where training window contamination is the strongest and threshold estimation is most prone to instability. AE-CFAR can still output a continuous target list under this working condition, indicating that its contamination suppression and edge perception strategies can effectively suppress the overall elevation of the threshold and edge false detections in a controlled electromagnetic environment.

Combining system indicators and statistical conclusions, in complex scenarios, AE-CFAR can reduce the false alarm rate by approximately 20% and improve detection accuracy by about 15% compared to other CFAR variants; at the same time, in the multi-target working condition at 150 m, the missed detection rate is reduced by more than 10%, and the end-side processing delay does not exceed 5 ms. This result has two implications: first, it verifies the stability of the algorithm under the pressure of "high concurrency + high dynamics"; second, it provides a comparable performance baseline for open road testing, so that changes in false/missed detections in subsequent road scenarios can be more attributed to environmental factors such as multipath, occlusion, and external interference, rather than instability in end-side computing or threshold mechanisms.

4.5 Real-Vehicle Testing in Complex Environments: Analysis of Multi-Vehicle Scenarios and Occlusions on Open Roads

4.5.1 Testing Scenarios and Data Collection Methods

The complex environment tests were conducted in an open urban road environment, which included multiple vehicles of different types traveling and parked along the roadside, with surrounding trees and buildings leading to significant environmental reflections and multipath effects. Testers carried data collection terminals and radar equipment to perform entry and exit tests, with the terminals displaying real-time distance and speed information corresponding to the vehicles; the on-site situation is shown in Figure 16. The challenges of this scenario for the detection link focus on three aspects: threshold estimation fluctuations caused by non-uniform backgrounds, short-term disappearance of targets due to vehicle occlusions, and the appearance of false peak structures in the RDM caused by multipath/interference. Therefore, the road tests mainly observed "the first detection distance of targets, the ability to recover from occlusions, and the continuity of output in concurrent multi-vehicle situations."

4.5.2 Results of Typical Working Conditions and Mechanistic Explanations

In road tests, the data collection terminal showed that targets entered the radar detection range at different times. As the vehicles approached, the target points on the

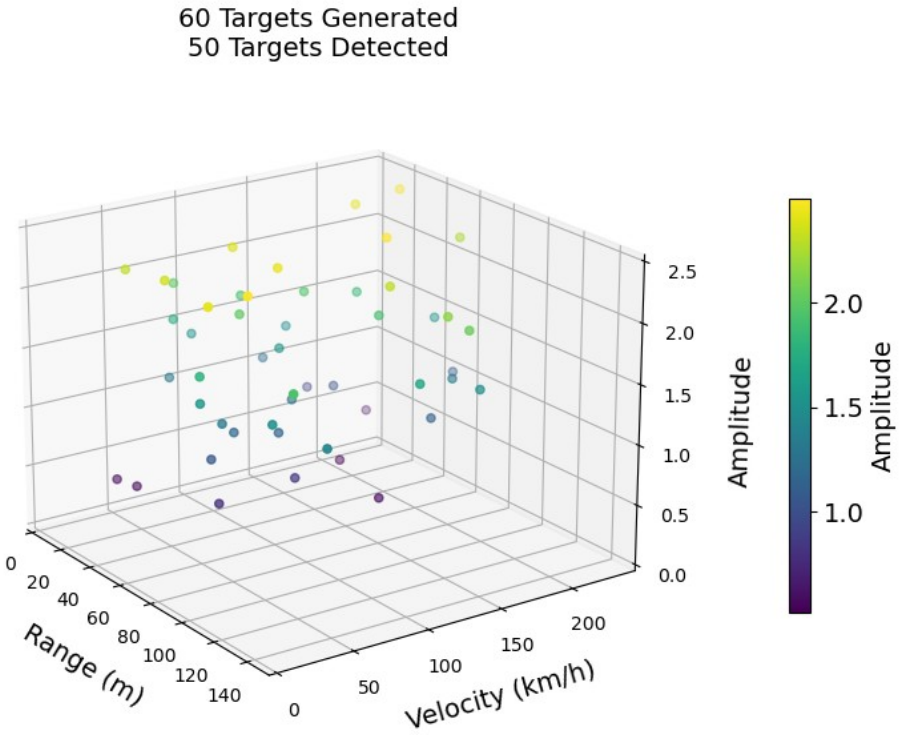


Fig. 15 Example of measured speed and distance results for 60 targets under anechoic chamber conditions

terminal interface were continuously updated, reflecting the system's real-time detection capability under the condition of multiple vehicles appearing simultaneously. In the occlusion scenario, the first detection distance of the red truck ahead was approximately 140 m; due to the white van closely following the red truck, causing complete occlusion, the white van was not detected at longer distances. When the vehicles approached to about 40 m and the white van moved left by approximately 0.5 m, the system re-detected the white van and output its distance and speed.

The above process can be used to explain the "recovery detection" characteristic of the 3T8R cascaded system in occlusion scenarios: when occlusion occurs, the target echo energy may drop below the threshold and be suppressed; when the target shifts laterally, its echo re-enters the main lobe coverage or the sidelobe interference is reduced, and the local peak structure in the RDM changes, allowing AE-CFAR to re-form stable candidate points in the neighborhood. On the other hand, 3T8R consistency fusion utilizes cross-channel/cross-time-slot redundancy, tending to retain target candidate points with consistent evidence, thus making it easier to quickly recover target output when occlusion is lifted or geometric relationships improve, without being long-term disturbed by occasional false peaks.

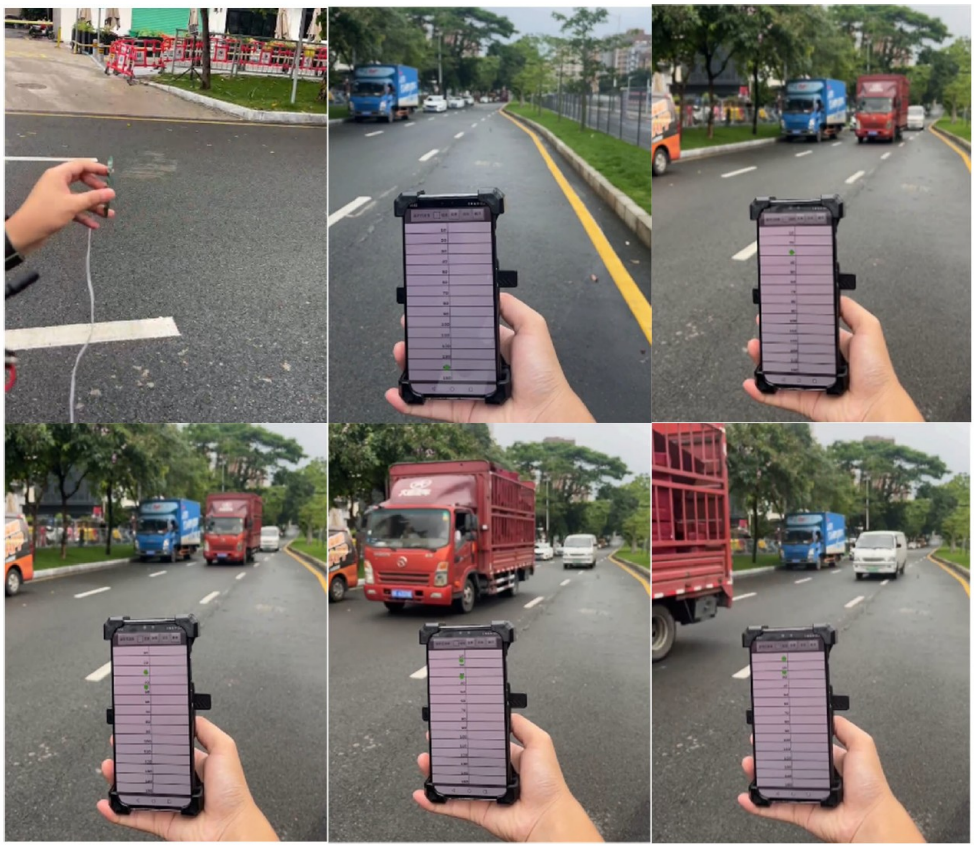


Fig. 16 Schematic diagram of the performance testing site in a complex environment

In addition, in urban roads with building reflections, the RDM background statistics often show segmental non-uniformity, and traditional threshold strategies are prone to false or missed detections in edge areas. The stable output of road tests indicates that edge-aware small-side selection can avoid threshold elevation by high-energy sides near background mutations, thereby improving the detectability of weak targets in complex backgrounds; at the same time, candidate point consistency screening can suppress isolated false peaks caused by residual interference or multipath, improving the continuity of the target list. The above observations complement the anechoic chamber tests: the anechoic chamber verified threshold stability under dense targets, while the open road verified engineering availability under occlusion, multipath, and complex background conditions.

5 Research Conclusions and Future Work

5.1 Research Conclusions

This paper addresses the multi-target detection requirements of automotive mmWave radar in complex traffic scenarios, focusing on the 24 GHz cascaded C-MIMO system and the AE-CFAR detection link, forming a complete technical closed loop from hardware architecture, edge-side implementation to system verification. At the system level, SRK2401 and SRK2402 are used to construct a cascaded transceiver front-end, forming a 3T8R observation structure with 3 TDM transmission time slots and 8 receiving channels, and realizing edge-side collaborative processing with MCU and DSP: MCU is responsible for front-end configuration, timing scheduling, and data flow management, while DSP is responsible for two-level FFT and detection calculations, thereby stably mapping multi-channel sampling data to RDM domain input and outputting candidate points and target lists. System index verification covers a detection range of 6–140 m and a speed measurement range of 14–225 km/h, with a distance resolution of ± 5 cm, a speed resolution of ± 0.7 km/h, and support for simultaneous tracking of up to 50 dynamic targets.

At the algorithm level, this paper constructs an AE-CFAR threshold detection method in the RDM domain, generating an adaptive threshold through truncated mean noise estimation for training window pollution suppression and distance-dimensional edge-aware small-side selection mechanism, while introducing candidate point consistency fusion combined with the 3T8R cascaded structure to suppress isolated false peaks caused by residual interference and occasional noise. In edge-side implementation, the algorithm adopts offline look-up tables for threshold coefficients and a fixed window scanning strategy to reduce computational overhead and stabilize processing load. Simulation and system-level test results show that the detection link has good threshold stability and continuity of candidate point output under conditions of dense multi-targets and enhanced noise; in comprehensive scenario statistics, the false alarm rate is reduced by approximately 20%, detection accuracy is improved by about 15%, the missed detection rate in 150 m multi-target scenarios is reduced by more than 10%, and the edge-side processing delay does not exceed 5 ms.

5.2 Research Limitations

The research in this paper still has several limitations. First, this paper focuses on RDM domain candidate point detection and consistency fusion, and the angular domain processing part mainly provides interfaces and implementation paths, without systematic evaluation of angular accuracy, angular resolution, and 3D positioning effects in the experimental chapter. Second, to meet edge-side real-time requirements, AE-CFAR adopts a robust estimation strategy with fixed windows and fixed culling ratios, but under extremely dense traffic or strong mutual interference conditions, the optimal values of window size, culling ratio, and consistency threshold may need further adaptation as scenarios change. Third, complex road tests verify system availability through typical event chains, but lack stricter truth alignment methods and error statistical systems, thus limiting quantitative conclusions on distance/speed error distributions and confidence intervals.

5.3 Future Work

To address the above limitations, future work will proceed in the following aspects. First, improve cascaded channel calibration and angular domain processing closed loop. Based on the existing 3T8R cascaded structure, establish an integrated calibration process for cross-chip amplitude-phase and group delay, conduct beamforming or DOA estimation experiments combined with virtual array observations after TDM reconstruction, and form an end-to-end evaluation system of "detection–angular domain separation–target list stability." Second, enhance interference and multipath robustness. While retaining the AE-CFAR threshold adaptive mechanism, study the structural characteristics of interference fringes and multipath false peaks in the RDM domain, establish discrimination and suppression strategies linked with consistency fusion, and reduce the probability of residual interference being continuously falsely detected as real targets. Third, construct lightweight scenario discrimination and parameter adaptation mechanisms, forming online updatable configuration strategies for training window size, culling ratio, edge threshold, and consistency threshold, to maintain stable false alarm control and weak target detection capabilities under different traffic densities and interference intensities, while ensuring controllable computational load on the DSP side. Fourth, expand the test system and form a statistical truth alignment method, introduce high-precision positioning and speed measurement means to establish error distributions and stability indicators, and supplement scenario tests such as rain and fog, night, tunnels, and dense interference to support more comprehensive engineering evaluation and large-scale deployment.

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Economy, Industry and Education (National Aeronautical Industry Teaching Steering Committee), the 33rd Golden Bridge Engineering Project of Deyang City, and the Sichuan Aerospace Vocational College Research Fund.

References

[1] Kong, H., *et al.*: A survey of mmwave radar-based sensing in autonomous vehicles, smart homes and industry. IEEE Communications Surveys & Tutorials **27**(1), 463–508 (2025) https://doi.org/10.1109/COMST.2024.3409556

[2] Yan, B., Roberts, I.P.: Advancements in millimeter-wave radar technologies for automotive systems: A signal processing perspective. Electronics **14**(7), 1436 (2025) https://doi.org/10.3390/electronics14071436

[3] Zhang, K., Meng, X., Wang, Q.: A review of recent advancements and applications of 4d millimeter-wave radar in smart highways. Urban Lifeline **3**(15) (2025) https://doi.org/10.1007/s44285-025-00048-1

[4] Yuan, S., Fioranelli, F., Yarovoy, A.: High-resolution imaging algorithms for automotive radar: challenges in real driving scenarios. IEEE Aerospace and Electronic Systems Magazine **40**(7), 30–43 (2025) https://doi.org/10.1109/MAES.2025.3550301

[5] Fei, T., Becker, A., Gardill, M.: Interference mitigation in automotive radar systems: A current state survey and future trends. IEEE Transactions on Intelligent Transportation Systems **27**(1), 53–80 (2026) https://doi.org/10.1109/TITS.2025.3629041

[6] Wang, J.: Cfar-based interference mitigation for fmcw automotive radar systems. IEEE Transactions on Intelligent Transportation Systems **23**(8), 12229–12238 (2022) https://doi.org/10.1109/TITS.2021.3111514

[7] Wang, J., *et al.*: Interference mitigation for automotive fmcw radar based on contrastive learning with dilated convolution. IEEE Transactions on Intelligent Transportation Systems **25**(1), 545–558 (2024) https://doi.org/10.1109/TITS.2023.3306576

[8] Xu, Z., Wei, S.: Fmcw radar system interference mitigation based on time-domain signal reconstruction. Sensors **23**(16), 7113 (2023) https://doi.org/10.3390/s23167113

[9] Liu, S., *et al.*: Design and performance validation of cwt-mca based interference mitigation for automotive radars. Digital Signal Processing **153**, 104644 (2024) https://doi.org/10.1016/j.dsp.2024.104644

[10] Li, X., *et al.*: Signal processing for tdm mimo fmcw millimeter-wave radar sensors. IEEE Access **9**, 167959–167971 (2021) https://doi.org/10.1109/ACCESS.2021.

3137387

[11] Gennarelli, G., *et al.*: 24 ghz fmcw mimo radar for marine target localization: A feasibility study. IEEE Access **10**, 68240–68256 (2022) https://doi.org/10.1109/ACCESS.2022.3186052

[12] Zhou, R., *et al.*: A two-chip cascaded fmcw radar for 2d angle estimation. In: 2021 IEEE MTT-S International Wireless Symposium (IWS), pp. 1–3 (2021). https://doi.org/10.1109/IWS52775.2021.9499433 . https://doi.org/10.1109/IWS52775.2021.9499433

[13] Kwiatkowski, P., *et al.*: Combining 77–81 ghz mimo fmcw radar with frequency-steered antennas: a case study for 3d target localization. International Journal of Microwave and Wireless Technologies **16**(5), 819–827 (2024) https://doi.org/10.1017/S1759078724000254

[14] Zheng, K., *et al.*: Enhancing mmwave radar sensing using a phased-mimo radar platform (hybradar). In: Proceedings of the 22nd Annual International Conference on Mobile Systems, Applications and Services (MobiSys '24), pp. 56–69 (2024). https://doi.org/10.1145/3643832.3661865 . https://doi.org/10.1145/3643832.3661865

[15] Benyahia, Z., *et al.*: High-resolution hybrid tdm-cdm mimo automotive radar. e-Prime - Advances in Electrical Engineering, Electronics and Energy **11**, 100897 (2025) https://doi.org/10.1016/j.prime.2025.100897

[16] Yang, B., Zhang, H.: A cfar algorithm based on monte carlo method for millimeter-wave radar road traffic target detection. Remote Sensing **14**(8), 1779 (2022) https://doi.org/10.3390/rs14081779

[17] Tian, F., *et al.*: Research on target detection algorithm for complex traffic scenes based on advi-cfar. Electronics **14**(7), 1474 (2025) https://doi.org/10.3390/electronics14071474

[18] Rihan, M.Y., Nossair, Z.B., *et al.*: An improved cfar algorithm for multiple environmental conditions. Signal, Image and Video Processing **18**, 3383–3393 (2024) https://doi.org/10.1007/s11760-024-03001-x

[19] Yang, X., Xiu, C.: Improved adaptive constant false alarm rate detector based on fuzzy theory for multiple-target scenario. Applied Sciences **15**(12), 6693 (2025) https://doi.org/10.3390/app15126693

[20] Liu, X., *et al.*: Multi-fold high-order cumulants based cfar detector for radar weak target detection. Digital Signal Processing **139**, 104076 (2023) https://doi.org/10.1016/j.dsp.2023.104076

[21] Li, J., *et al.*: A novel copula-based multi-feature cfar framework for multi-feature detection. Remote Sensing **17**(13), 2299 (2025) https://doi.org/10.3390/rs17132299

[22] Yang, H., et al.: From spatial domain to temporal domain: Unleashing the capability of cfar for mmwave point cloud generation. ACM Transactions on Sensor Networks **9**(2) (2025) https://doi.org/10.1145/3729465

[23] Kang, S.-W., *et al.*: Autoencoder-based target detection in automotive mimo fmcw radar system. Sensors **22**(15), 5552 (2022) https://doi.org/10.3390/s22155552

[24] Zhang, Y., *et al.*: Random time division multiplexing based mimo radar processing with tensor completion approach. Sensors **23**(10), 4756 (2023) https://doi.org/10.3390/s23104756

[25] Wang, D., *et al.*: Grouped target tracking and seamless people counting with a 24-ghz mimo fmcw. IEEE Transactions on Radar Systems **3**, 1298–1308 (2025) https://doi.org/10.1109/TRS.2025.3609436

[26] Inc., S.S.: SRK2401: 24 GHz 2T4R FMCW Radar Transceiver Data Sheet, Rev. A1.6. Datasheet (2025). https://atta.szlcsc.com/upload/public/pdf/source/20250619/27F7F44B88D3946C3201A8607356DED4.pdf

[27] Li, B., Sangalang, R.G.B.: A sub-12 db c-mimo array design for millimeter-wave radar applications. In: 2024 IEEE 16th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM), pp. 1–6 (2024). https://doi.org/10.1109/HNICEM64917.2024.11258676 . https://doi.org/10.1109/HNICEM64917.2024.11258676